

Question Paper

Exam Date & Time: 03-May-2025 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FOURTH SEMESTER B.TECH. (COMPUTER SCIENCE & ENGINEERING) DEGREE EXAMINATIONS - MAY 2025
SUBJECT: CSE 2222/CSE_2252 - DESIGN AND ANALYSIS OF ALGORITHMS

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Consider the following algorithm. (4)
- ```
ALGORITHM Magic ($A[0..n-1]$)
 $dmagic \leftarrow \infty$
for $i \leftarrow 0$ to $n-1$ do
 for $j \leftarrow 0$ to $n-1$ do
 if $i \neq j$ and $|A[i] - A[j]| < dmagic$
 $dmagic \leftarrow |A[i] - A[j]|$
return $dmagic$
```
- i) What does this algorithm compute?  
ii) What is its basic operation?  
iii) How many times is the basic operation executed?  
iv) How to improve the efficiency of this algorithm?
- 1B) Write a recursive algorithm to count the number digits in octal equivalent of a decimal number. Set up and solve the recurrence relation. (3)
- 1C) Solve the following recurrence relation using backward substitution method. (3)
- i)  $M(n) = 2M(n-1) + 1$  for  $n > 1$ ,  $M(1)=1$ .  
ii)  $T(n) = T(n/2) + n$  for  $n > 1$ ,  $T(1) = 1$
- 2A) Compare Depth First Search with Breadth First Search graph traversal algorithms. (3)
- 2B) Apply topological sorting using DFS on the graphs shown in Figure 2B. How many different solutions possible in each graph? (3)

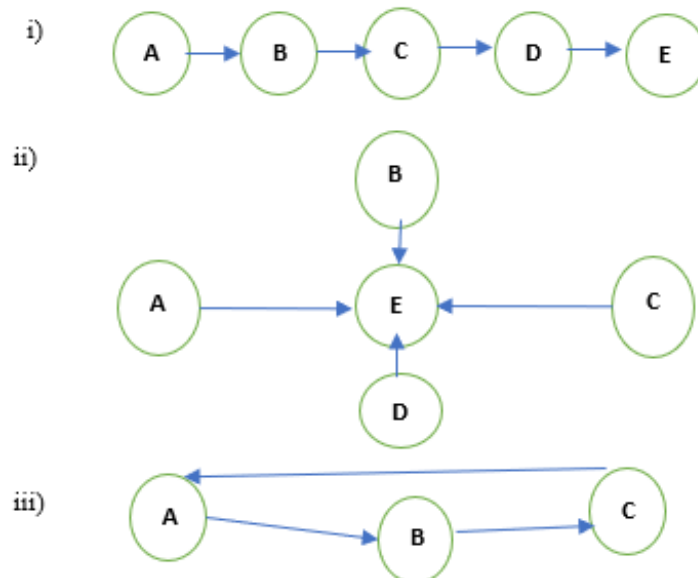


Figure 2B

- 2C) Design selection sort algorithm. Find its time complexity. Sort the following elements using selection sort (4)  
**88, 44, 67, 89, 28, 33, 16**
- 3A) Analyse the best-case and worst-case time complexity of quicksort algorithm using Hoare partition, considering key comparison as the basic operation. Apply the same Quicksort algorithm to the following set of elements **43, 12, 76, 16, 10, 25, 70, 33** to arrange in ascending order and draw the tree of recursive calls indicating the different partition positions (starting index 0) (4)
- 3B) What is an AVL tree? Construct AVL tree for the list **12, 13, 14, 15, 10, 9, 6, 20, 18, 19** by successive insertion method starting from empty tree. Show all stages. (2)
- 3C) Sort the list: 40, 32, 65, 98, 18, 62, 30, 75 in ascending order using heapsort by clearly showing top-down heap construction and sorting stages (4)
- 4A) Given a knapsack with a maximum weight capacity of 6, and the following items with their respective weights and profits: (4)
- Item 1: weight 3, profit \$25
  - Item 2: weight 2, profit \$20
  - Item 3: weight 2, profit \$15
  - Item 4: weight 4, profit \$40
  - Item 5: weight 5, profit \$50
- Determine the combination of items that maximizes the total profit while staying within the knapsack's weight limit, using a memory function (memorization) approach. What is the optimal solution, and which items are included?
- 4B) Apply Horspool's algorithm to search for the pattern **MATCH** in the given text **HORSPOOL-PATTERN MATCHING-ALGORITHM** by constructing the bad shift table, clearly showing all the stages and find total number of character comparison. (3)
- 4C) Apply Floyd's algorithm to the graph shown in Figure 4C and find the shortest path distance between every pair of vertices. (3)

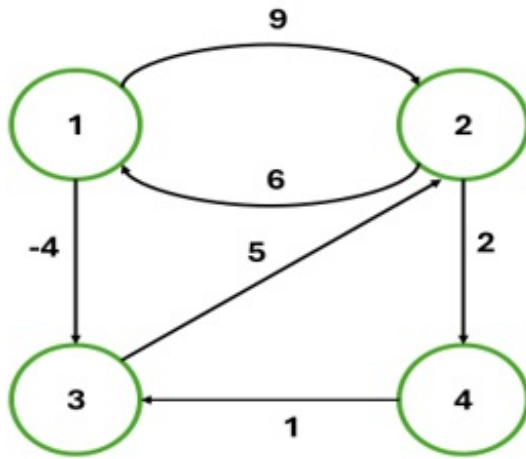


Figure 4C

- 5A) Solve the following subset sum problem by using the backtracking method,  $S = \{3, 34, 4, 12, 5, 2\}$  and  $d = 9$ . Write a complete state space diagram and compute all the solutions. (3)
- 5B) Consider an application that compresses text files before they are sent over a network. The application uses Huffman coding for compression. The frequency of characters in a particular file you are testing is as follows: A (5 times), B (9 times), C (12 times), D (13 times), E (16 times), and F (45 times). Write the general procedure of constructing a Huffman tree for the given character set. Illustrate the construction of the Huffman Tree for the above character set and find the Huffman Code for each character. Calculate the compression ratio. (4)
- 5C) Apply Prim's Algorithm to determine the minimum spanning tree for the network given below in Figure 5C. Begin from the S as the starting vertex. Explain how Prim's Algorithm selects the edges and updates the potential minimum costs to connect all the areas with minimal expense. (3)

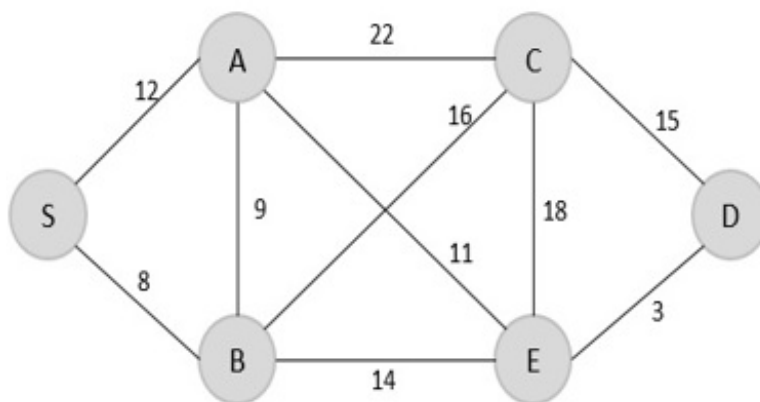


Figure: 5C

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| Event Name                                                                                   |                | Subject                                    |       |
|----------------------------------------------------------------------------------------------|----------------|--------------------------------------------|-------|
| MIT MPL and BLR- BTech-M Sc-MCA - II-IV and VI Sem - End Semester Examination - Apr-May 2025 |                | CSE-2252:Design and Analysis of Algorithms |       |
| #                                                                                            | Course Outcome | Description                                | Marks |

No records found / CO Level Description not defined

\* **NA** - CO Not Selected

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| #                                                                                            | Level | No of Questions                            | Marks (%) |

No records found / Blooms Level Description not defined